

Maulana Azad National Institute of Technology Bhopal-462003

End Term Examination, December-2022

B.Tech.

Semester-III

Branch: CE/Chemical/ECE/EE/ME/MME

Subject: Mathematics-III

Code: MTH-231

Max. Marks: 50

Time: 3 Hours

Note: All questions are compulsory. Usual notations are used. The answer should be correct to four decimal places (4D).

S.No.	Question	Marks																						
1(a)	Find the real root of the equation $x^2 + 4\sin x = 0$ by using the Newton-Raphson method.	5																						
1(b)	Apply Gauss Seidel method to solve the following system of equations $8x - 3y + 2z = 20$, $6x + 3y + 12z = 35$, $4x + 11y - z = 33$.	5																						
2(a)	Following table gives the population of a town during the last six censuses. Estimate using any suitable interpolation formula, the increase in the population during the period from 1946 to 1948. <table><tr><td>Year (x)</td><td>1911</td><td>1921</td><td>1931</td><td>1941</td><td>1951</td><td>1961</td></tr><tr><td>Population (in 1000's)</td><td>12</td><td>15</td><td>20</td><td>27</td><td>39</td><td>52</td></tr></table>	Year (x)	1911	1921	1931	1941	1951	1961	Population (in 1000's)	12	15	20	27	39	52	5								
Year (x)	1911	1921	1931	1941	1951	1961																		
Population (in 1000's)	12	15	20	27	39	52																		
2(b)	Use the divided difference method to find the interpolating polynomial for the given data: <table><tr><td>x</td><td>4</td><td>5</td><td>7</td><td>10</td><td>11</td><td>13</td></tr><tr><td>f(x)</td><td>48</td><td>100</td><td>294</td><td>900</td><td>1210</td><td>2028</td></tr></table> Also, find the value of the polynomial at 2, 8, 9 and 15.	x	4	5	7	10	11	13	f(x)	48	100	294	900	1210	2028	5								
x	4	5	7	10	11	13																		
f(x)	48	100	294	900	1210	2028																		
3(a)	A slider in a machine moves along a fixed straight rod. Its distance x in cm along the rod is given below for various values of the time t in seconds. Find the velocity of the slider and its acceleration when $t = 0.3$ seconds. <table><tr><td>T</td><td>0</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td><td>0.6</td></tr><tr><td>x</td><td>30.13</td><td>31.62</td><td>32.87</td><td>33.64</td><td>33.95</td><td>33.81</td><td>33.24</td></tr></table>	T	0	0.1	0.2	0.3	0.4	0.5	0.6	x	30.13	31.62	32.87	33.64	33.95	33.81	33.24	5						
T	0	0.1	0.2	0.3	0.4	0.5	0.6																	
x	30.13	31.62	32.87	33.64	33.95	33.81	33.24																	
3(b)	The velocity v (km/min) of a car which starts from the rest is given at the fixed intervals of time t (min) as follows. Estimate approximately the distance covered in 20 minutes by using the Simpson's 3/8 rule. <table><tr><td>T</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td><td>16</td><td>18</td><td>20</td></tr><tr><td>x</td><td>10</td><td>18</td><td>25</td><td>29</td><td>32</td><td>20</td><td>11</td><td>5</td><td>2</td><td>0</td></tr></table>	T	2	4	6	8	10	12	14	16	18	20	x	10	18	25	29	32	20	11	5	2	0	5
T	2	4	6	8	10	12	14	16	18	20														
x	10	18	25	29	32	20	11	5	2	0														

4(a)	Use modified Euler's method to find the approximate value of y when $x = 1.2$ and 1.4 for $\frac{dy}{dx} = \frac{2y}{x} + x^3, y(1) = 0.5$ with step length 0.2 .	5																								
4(b)	Solve the equation $\frac{dy}{dx} = y - x, y(0) = 2$ by using the Runge-Kutta method of fourth order to find the value of $y(0.1)$ and $y(0.2)$.	5																								
5(a)	Calculate the correlation coefficient and the line of regression of X on Y if the marks of two subjects X and Y are as: <table><tr><td>X</td><td>80</td><td>45</td><td>55</td><td>56</td><td>58</td><td>60</td><td>65</td><td>68</td><td>70</td><td>75</td><td>85</td></tr><tr><td>Y</td><td>82</td><td>56</td><td>50</td><td>48</td><td>60</td><td>62</td><td>64</td><td>65</td><td>70</td><td>74</td><td>90</td></tr></table>	X	80	45	55	56	58	60	65	68	70	75	85	Y	82	56	50	48	60	62	64	65	70	74	90	4
X	80	45	55	56	58	60	65	68	70	75	85															
Y	82	56	50	48	60	62	64	65	70	74	90															
5(b)	In a precision bombing attack there is a 50% chance that any one bomb will strike the target. Two direct hits are required to destroy the target completely. How many bombs must be dropped to give a 99% chance or better of completely destroying the target ?	3																								
5(c)	Find the best-fit values of a and b so that $y = a + b x$ fits the data given in the table <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>3.3</td><td>4.5</td><td>6.3</td></tr></table>	x	0	1	2	3	4	y	1	1.8	3.3	4.5	6.3	3												
x	0	1	2	3	4																					
y	1	1.8	3.3	4.5	6.3																					